
Can This Agent Even Do That?

A Decidable Goal-Achievability Type Discipline for LLM-Synthesized Agent Skills

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Abstract

1 Autonomous agents are increasingly configured by *skills*: natural-language speci-
2 fications that an LLM drafts from a user’s stated intent and then executes, often
3 across several cooperating agents. Such drafts fail expensively at run time on
4 plans that were impossible from the start — a step that invokes a tool the agent
5 does not have, a goal conjunct nothing can establish, a budget no run can meet,
6 or a multi-agent handoff that deadlocks. We present a *skill compiler* that decides,
7 before execution and without an LLM judge, whether a skill’s *goal is achievable at*
8 *all*. The compiler fuses two mature disciplines not previously combined for this
9 purpose: capability-guarded reachability from automated planning, and deadlock-
10 freedom from multiparty session types (MPST), decided *directly* over session
11 configurations and a goal-marked global type synthesized from intent, following
12 the synthetic MPST discipline of checking a session directly against a global pro-
13 tocol [1, 2]. The LLM that compacts prose into a formal pack of preconditions,
14 effects, and goals is treated as *untrusted*; the checker is specified by a Coq-verified
15 refutation-sound abstraction theorem. Relative to the declared pack, an *impossible*
16 verdict is never wrong; whether the pack is faithful to the prose is a separately
17 deferred obligation. The positive verdict remains deliberately incomplete, with
18 payload faithfulness and intent fidelity handed respectively to a runtime monitor
19 and a human checkpoint. We also establish subject reduction and session fidelity
20 for the direct discipline: the single-label communication fragment with bystander
21 interleaving is mechanized, while world-changing action interleaving is proved
22 on paper under explicit goal-stability and effect-commutativity hypotheses. We
23 give the syntax, labelled operational semantics, typing and achievability judgments,
24 and the metatheory: refutation, tolerance and capability-monotonicity soundness;
25 operational correspondence; decidability within a static-topology fragment; and an
26 undecidability boundary for an extension with dynamic participant spawning. An
27 implementation and reproducible corpus exercise the canonical failure modes, the
28 fragment boundary, and the two deferred edges.

29 1 Introduction

30 A modern agent is rarely a single model call, but neither is it simply a model. Around the model
31 sits a *harness*: the runtime infrastructure that fixes the execution environment, the tool interface, the
32 context, the orchestration and the verification, and on which reliability depends at least as much as on
33 the base model [3, 4]. A *skill* is one increasingly common way of writing part of that harness down —
34 a natural-language artifact, a folder of instructions the agent discovers and loads on demand [5], and
35 increasingly drafted by an LLM from a user’s stated goal [6, 7] — that *harnesses* a general-purpose

agent toward a particular outcome, naming the tools it may use, prescribing the plan it should follow and, when several agents are involved, fixing who must talk to whom. A skill is behavioural guidance layered over an agent, not a substitute for it: the agent retains its latitude to interpret that guidance at run time. The appeal is that a non-expert can say *what* they want and have the system assemble the *how*. The question the arrangement leaves open — and the question this paper answers — is how anyone knows that the guidance is good enough to achieve what it claims.

Today that answer arrives only by running the skill and reading the wreckage afterwards [4]. Across seven state-of-the-art open-source multi-agent systems, measured failure rates range from 41% to 86.7% [8]; on one of them a specification-level intervention alone — adding a task-objective verification step, with the underlying model held fixed — yields a 15.6% improvement in task success [8, Appendix H], and trace-driven repair of the harness alone lifts task completion by 6.3 to 18.4 points across four agent benchmarks [4]. A substantial share of the defect therefore lies in the guidance rather than in the model. Part of that share is worse than unlucky: the assembled how was *impossible* before the first token was spent, and nothing noticed until execution had already taken actions and, in the multi-agent case, deadlocked halfway.

The failures are structural and recurring. A plan *books a flight and emails a confirmation* although no email tool is in scope (a hallucinated capability). A plan pursues *a flight under \$500* when the only booking tool returns premium fares (an unsatisfiable refinement). A planner waits for a worker’s result while the worker waits for the planner’s answer, each listening for a message the other never sends (a coordination deadlock). In every case the goal was unreachable *a priori*, and a cheap static check could have said so.

This paper asks a deliberately narrow question and answers it with a guarantee: **given an LLM-drafted skill, can its goal be achieved at all?** We are not (here) verifying that every step is safe, nor that the agent behaves correctly on every input. We decide *achievability*, tolerantly — small details, detours, and payload variation must not raise a false alarm — and *soundly for refutation*: when the compiler says a goal is impossible, it is right.

1.1 The idea, and the trust boundary

The architecture has two halves separated by a trust boundary. An LLM performs *compaction*: it reads the prose skill and distills it into a formal *pack* — capabilities annotated with preconditions and effects, a goal-marked global protocol, and an initial state. This step is *untrusted*; the LLM may misread the prose. A deterministic *checker* then decides achievability over the pack, and the checker is mechanically proved sound. The division is what makes the system honest: a faulty compaction can only ever cause a false *achievable* (caught downstream), never a false *impossible* about the pack it received (Figure 1).

Crucially, the semantic gap between *what the user meant* and *what was formalized* is not eliminated — it is *relocated and concentrated* into one inspectable checkpoint at compaction time, rather than scattered, unobserved, across every run. That relocation is the central design move.

1.2 Contributions

1. **A goal-achievability type discipline** (Section 3 and Appendices A–C) that fuses capability-guarded reachability (planning) with deadlock-freedom (multiparty session types) over a goal-marked global type synthesized from intent. Following synthetic MPST [1, 2], the declarative discipline types whole session configurations directly; our contribution is to add the guarded world, goal observation, and refutation-sound/achievement-incomplete verification asymmetry for LLM-drafted agent skills.
2. **A mechanized soundness specification** (Appendix D) in Coq: a refutation-sound abstraction theorem parameterized by simulation hypotheses, tolerance soundness (coarsening the abstraction preserves refutation), and capability monotonicity (more tools never make a goal impossible) — all axiom-free — together with an *operational-correspondence* metatheory (subject reduction and session fidelity) for the direct judgment over a labelled transition system. Full bystander interleaving is mechanized for single-label communication; the world-changing action cases are paper proofs under explicit side conditions.

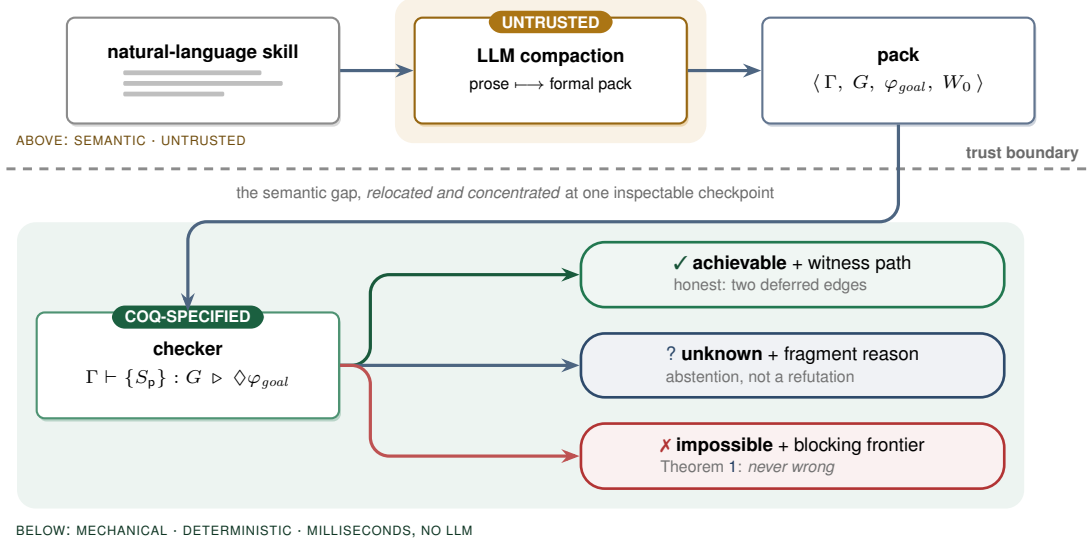


Figure 1: The trust boundary. An untrusted LLM compacts prose into a formal pack; below the line, a deterministic checker governed by a Coq-verified abstraction theorem decides achievability over the pack — the pack is the only artifact that crosses. The verdicts are asymmetric: *impossible* is sound (Theorem 1), *achievable* is honest about its deferred obligations (Section 3.1).

3. **A decidability boundary result** (Appendix D): achievability is decidable in a static-topology fragment; an explicit extension with unbounded participant spawning crosses to undecidability.
4. **An implementation and evaluation** (Sections 4–5): a checker, an LLM-compaction front-end with a schema gate, and a corpus spanning the canonical failure modes. Results match the theory: zero false impossibilities; every false achievability is one of the two deferred residues.

2 Overview by Example

Consider a one-agent skill whose goal is *a flight is booked and a confirmation is sent*. The author writes prose; the LLM compacts it. Two compactions differ only in whether an email tool is in scope.

$$\varphi_{goal} = \text{booked} \wedge \text{confirmation_sent}$$

capability	precondition	effect
<i>variant A</i> — no email tool in scope		
search	$\top$	$Add\{\text{searched}\}$
filter	searched	$Add\{\text{filtered}\}$
book	filtered	$Add\{\text{booked}\}$
<i>variant B</i> — variant A, plus:		
email	booked	$Add\{\text{confirmation_sent}\}$

Under STRIPS *frame semantics* — a predicate is false unless an effect makes it true — variant A has no action establishing `confirmation_sent`, so no reachable state satisfies the goal: the checker returns *impossible* and points at the missing establishment. Variant B reaches the goal along `search · filter · book · email`, returned as a witness. The same example is mechanized in Appendix D (the concrete instance `FlightInstance`): variant A is *provably* unachievable, yet a booking is still reachable — the refutation is about the missing tool, not a dead protocol.

Tolerance is essential, and is where a naive reachability check would misfire. Extra status messages, a clarifying detour, or payload variation must not refute the goal. We obtain this from three orthogonal relaxations made precise in Appendices A–C: *existential may-reachability* (a single good path

107 suffices), *session subtyping* (interface slack in the safe direction, folded directly into the single
 108 conformance rule rather than a separate relation), and *goal-relevant abstraction* (payload detail
 109 collapsed before checking).

110 3 Verification Model and Signals

111 **Declared pack.** The checker consumes $P = \langle \Gamma, G, \varphi_{goal}, W_0 \rangle$: capability declarations Γ with
 112 guards and STRIPS-style effects; a goal-marked global protocol G (formally, a *global type*; Ap-
 113 pendix A); a logical goal φ_{goal} ; and an initial abstract world W_0 . The LLM that compacts prose into
 114 P is outside the trusted core. A deterministic schema gate first enforces the decidable fragment: finite
 115 static roles, guarded recursion, finite-domain booleans, and quantifier-free linear integer arithmetic.
 116 The formal syntax and labelled semantics are given in Appendices A–B.

117 **Heterogeneous verification signals.** Achievability is not reduced to a learned score. The judgment
 118 composes signals that live on different substrates:

Signal	Formal role	Evidence returned
tool availability and effects coordination structure	capability guards; world transition direct global conformance	missing capability or witness action offending handoff or conformant proto- col
goal and cost refinements	logical reachability; QF-LIA	unsatisfied conjunct, blocked guard, or model
human intent review runtime payload checks	validates prose-to-pack fidelity deferred monitor obligation	accepted or revised declared goal blame-bearing execution event

120 The first three signals are composed before execution. The latter two are explicit obligations rather
 121 than assumptions silently folded into the static verdict.

122 **Decision procedure.** After the schema gate, the checker (i) rejects references to undeclared capa-
 123 bilities and goal atoms with no establisher; (ii) checks declared role behaviors against G , requiring
 124 exact sender choices while allowing receiver-side branch supersets; and (iii) explores existential pro-
 125 tocol/world successors from W_0 , using Z3 for guards and refinements. An unsatisfied goal marker is
 126 a blocking checkpoint. Solver timeouts and packs outside the static fragment cannot justify refutation.
 127 The interface is therefore tri-valued:

IMPOSSIBLE(F)	a sound blocking frontier F ,
ACHIEVABLE(π, O)	a witness path π and deferred obligations O ,
UNKNOWN(r)	an abstention with fragment reason r .

128 Verdicts also carry the pack digest, checker semantics, and artifact version so that feedback remains
 129 attributable to the exact declaration checked.

130 **Guarantees.** Let abs map concrete protocol/world configurations to abstract configurations, and
 131 assume step and goal simulation for the declared pack (Appendix D). If no abstract run reaches
 132 φ_{goal} , no concrete run represented by that pack reaches the concrete goal (Theorem 1). Coarsening
 133 α and adding capabilities cannot create a false refutation (Theorems 2–3). Direct conformance has
 134 subject-reduction and session-fidelity results (Theorems 4–5); the Coq artifact covers single-label
 135 communication with bystander interleaving, while the paper states explicit stability and commutativity
 136 assumptions for world-changing interleavings. The static fragment is decidable; unbounded dynamic
 137 spawning is an explicit undecidable extension, for which the implementation abstains. Full rules,
 138 assumptions, proofs, and mechanization scope appear in Appendix D.

139 3.1 Structured feedback and human oversight

140 The blocking frontier and witness are inspectable, heterogeneous feedback rather than a scalar reward.
 141 A bounded repair adapter may ask the untrusted LLM to fix a non-projectable protocol, but it may
 142 neither invent capabilities nor weaken the goal, and every revision is rechecked deterministically.
 143 This realizes a small agent-verifies-agent loop while keeping the human checkpoint at the irreducible
 144 intent-fidelity stage. Automated prompt or scaffold search is an application of these verdicts, not an
 145 empirical contribution claimed here.

146 4 Implementation

147 The executable decision path is approximately 1.45K lines of Python, including the checker, pack gate,
 148 formula layer, and session adapter. The compaction front-end issues the artifact prompt to an LLM
 149 and passes the result through a deterministic *schema gate* before the checker sees it, so malformed
 150 packs are rejected without crashing the trusted core. The Coq artifact contains three developments and
 151 two assumption-audit harnesses: reachability soundness core (Theorems 1–3 plus `FlightInstance`,
 152 ~ 230 lines); the direct-typing core (the head-move action safety plus `HandoffInstance`, ~ 310
 153 lines); and the operational-correspondence core (Theorems 4–5 for the single-label communication
 154 fragment with full interleaving, ~ 250 lines); all three are checked in CI under pinned Coq 8.18.0
 155 with no axioms, verified by `Print Assumptions`. The checker implements the corresponding
 156 abstractions from Appendices A–C: STRIPS frame semantics for the world, existential search over
 157 the protocol control structure, and Z3 for guard/goal satisfiability and arithmetic refinements. On
 158 success it returns the witness path; on refutation, the blocking frontier (missing capability, unsatisfiable
 159 guard, unestablished goal conjunct, or non-projectable handoff). The optional LLM front-end may
 160 use a `NON_PROJECTABLE` counterexample for one bounded, structure-only repair round; it may not
 161 invent capabilities or weaken the goal, and the deterministic checker remains the final referee. This is
 162 a small reflective-verification loop, not a general scaffold-search result.

163 The conformance step ($G \vdash (\mathbb{M}; W)$, Section C.2) is approximated *algorithmically* in `session.py`.
 164 The adapter first derives the behavior expected of each role (projection), reconciles choices that a
 165 bystander cannot observe (merge), and then compares each declared role with that expectation (direct
 166 conformance). The declarative judgment is what the checker must decide, projection-with-merge
 167 remains a sound and, we expect, complete decision procedure for it whenever G is realizable —
 168 that equivalence is asserted from the classical correspondence between projection and network-level
 169 conformance [12, 9], not itself mechanized here. The conformance adapter matches T-COMM: sender
 170 labels must equal the protocol labels, while a receiver may accept a superset. A separate generic
 171 Gay–Hole subtyping utility remains in the artifact but does not decide this premise. The verdict reason
 172 `NON_PROJECTABLE` names this algorithmic engine’s diagnostic; Section C.2 is the specification it is
 173 checked against.

174 The adapter also discharges the participant side conditions of T-COMM/T-ACT/T-GOAL: it computes
 175 $\text{prt}(G)$ and compares it with the roles the pack declares behaviour for. One direction refutes — a
 176 declared role outside $\text{prt}(G)$ has contract `End`, which nothing non-trivial refines. The other cannot: a
 177 participant the pack declares nothing for offers no behaviour to refute, yet the judgment of Section C.2
 178 ranges over a whole session with $\text{prt}(G) = \text{prt}(\mathbb{M})$, so the checker returns those roles alongside the
 179 verdict rather than leaving the assumption implicit.

180 5 Evaluation

181 We assembled a corpus of 15 skills spanning the canonical failure modes — hallucinated planning,
 182 retry-forever loops, coordination deadlock, refinement violations — with achievable controls (in-
 183 cluding detours and informed branches) and two deliberately *spurious* cases exercising the deferred
 184 edges. Each skill carries a ground-truth semantic label; the positive class is *achievable*. Evaluation is
 185 deterministic and CPU-only: it requires no training or live model inference, completes the 15-case
 186 matrix in under one second on a commodity laptop, and gives each individual Z3 query a 10-second
 187 timeout.

	predicted achievable	predicted impossible
truly achievable	TP = 6	FN = 0
truly impossible	FP = 2	TN = 7

189 Two claims, both confirmed. **Soundness:** FN = 0 — no achievable goal was wrongly refuted,
 190 the empirical face of Theorem 1 (achievability recall $6/6 = 100\%$). **Incompleteness, contained:**
 191 FP = 2, and both are exactly the planted spurious cases — one a false payload post-condition
 192 (Layer-C obligation), one an under-specified goal (intent edge). No structural failure was missed in
 193 this proof-of-concept corpus; this is evidence of coverage, not a proof that the two residues exhaust
 194 all empirical failure modes. Each refutation reason fires on its matching mode:

	Failure mode (source)	Verdict reason
	hallucinated planning — invokes a non-existent tool	MISSING_CAPABILITY
195	hallucinated planning — no establisher for a goal conjunct	GOAL_UNSAT
	retry-forever loop — guard never satisfiable	BLOCKED_GUARD
	coordination deadlock — unobserved choice / bad handoff	NON_PROJECTABLE
	refinement violation — budget unsatisfiable on every run	GOAL_UNSAT

196 **What the check costs.** With no model in the trusted path, the checker and the deterministic front-
 197 end spend *zero* tokens; only compaction spends any, once per skill *version* and linearly in the skill’s
 198 length, against a run it avoids that recurs per *invocation* and is quadratic in its turns. Under an explicit,
 199 conservative model of that run (Appendix E), compacting the 15-spec corpus costs 12.2K tokens
 200 and avoids $\sim 1.07\text{M}$ per round of invocations — $\sim 87\times$ leverage, unbounded deterministically — so
 201 break-even falls inside the first prevented run in every failure mode, while a skill that is *not* broken
 202 pays 2.9% of one successful run for the check, or nothing.

203 A separate six-case fragment suite exercises behavior absent from the headline matrix: tail-recursive
 204 retry and non-terminating control, dynamic spawning, protocol-independent refutation despite spawn-
 205 ing, and conformant versus non-conformant declared role behavior. It yields two ACHIEVABLE,
 206 three IMPOSSIBLE, and one UNKNOWN. The abstention is reported separately rather than counted
 207 as a false negative, because UNKNOWN makes no refutation claim. Together the suites contain 21
 208 declared packs; the 15-case matrix remains the only binary ground-truth matrix.

209 6 Related Work

210 **Prior-art note.** The claims below reflect the authors’ survey at submission time; the fusion’s novelty
 211 should be confirmed against a systematic search of the planning-meets-behavioural-types literature,
 212 which is dispersed across communities.

213 *Multipart session types* [12, 15, 9] supply deadlock-freedom via projection and subtyping [13,
 214 10]; our calculus and directed-choice global types follow the modern synchronous formulation
 215 of the Yoshida line [9–11]. We adopt the synthetic line [1, 2]: Section C.2 checks conformance
 216 directly against a whole session configuration rather than projected local types. Our direct rule
 217 corresponds to the conservative, plain-merge condition that bystanders behave uniformly across
 218 an unobserved choice; it does not subsume every protocol accepted by modern full merge. The
 219 new contribution is the world and goal layer: capability guards, effects, goal observation, and
 220 the asymmetric refutation guarantee are not modeled by classical or synthetic MPST. *Automated*
 221 *planning* [16] supplies capability-guarded goal reachability; we reuse its frame semantics but embed
 222 it in a multipart choreography. The contribution is the *fusion*, decided over an goal-marked global
 223 type synthesized by an LLM. Recent agent-verification work is adjacent but differently aimed:
 224 *provable coordination via message sequence charts* [17] projects a global choreography for LLM
 225 agents but targets coordination correctness, not goal achievability; its runtime-planning extension
 226 also establishes that LLM synthesis of a coordination protocol is itself prior art. *VeriGuard* [18]
 227 separates offline policy verification from online monitoring as a safety shield; *agent behavioural*
 228 *contracts* [19] provide probabilistic, runtime-enforced compliance and composition conditions for
 229 multi-agent chains, whereas our verdict is static and sound for refutation relative to the pack.

230 *Skill-bundle scanners and verified-skill catalogs* are complementary. NVIDIA’s SkillSpector [20] and
 231 Verified Agent Skills pipeline [21] address the same deployment object—portable agent skills—but
 232 ask a different question. SkillSpector is a pre-publication security control: it normalizes a skill bundle
 233 and runs deterministic scanners such as static pattern detectors, AST and taint analyzers, manifest-
 234 consistency checks, metadata-poisoning checks, supply-chain checks, and optional LLM-backed
 235 semantic analyzers. This is valuable admission control for skills, but it is not a compiler or type
 236 discipline in the sense used here. As published, SkillSpector does not aim to provide an operational
 237 semantics of goal-marked skills, a goal-reachability decision, or a refutation-soundness theorem. A
 238 SkillSpector-like pass belongs at the boundary of the compiler: before or alongside LLM compaction,
 239 it can vet the incoming SKILL.md, code, dependencies, and MCP metadata; after compaction, it
 240 can help justify the honesty of declared capabilities and least-privilege metadata. The type checker
 241 then consumes the sanitized formal pack and proves the narrower claim that the declared goal is not
 242 structurally impossible.

243 None of the above frames the static check as goal achievability over a declared world and goal-marked
244 protocol, nor gives the same refutation-sound/achievement-incomplete asymmetry.

245 7 Limitations and Future Work

- 246 • **Relative soundness.** Everything is proved about the declared Γ and S . If the prose lies (a
247 “read-only” tool that writes), the checker verifies a fiction; honest-declaration is a separate
248 obligation, of which this makes only the interaction slice decidable. A practical compiler
249 should therefore include a skill-bundle security pass—for example a SkillSpector-like
250 scanner—to inspect SKILL.md, scripts, dependencies, manifests, and permission metadata
251 before the formal pack is trusted by the achievability checker.
- 252 • **Static topology for totality.** Dynamic spawning drops the procedure to a semi-decision
253 (Proposition 2); a practical system answers UNKNOWN rather than guessing.
- 254 • **Trusted abstraction.** A coarse α silently widens tolerance (more false achievable), never
255 breaking Theorem 1 but weakening usefulness.
- 256 • **Engineering gaps.** The mechanization covers the generic refutation-sound abstraction
257 theorem and concrete examples, head-move action safety, and subject reduction/session
258 fidelity with full bystander interleaving for the single-label communication model, all axiom-
259 free. The executable checker is schema-gated and tested against that specification, but the
260 STEP-SIM/GOAL-SIM instantiation for its exact symbolic state is not mechanized. The Coq
261 correspondence model also does not yet combine observable goal labels, participant match-
262 ing, multi-branch choice, and world-changing interleaving in one development. Still open
263 are that faithful mechanization; a head-normalization/permutation bridge from interleaved
264 session runs to the head-ordered reachability search; equivalence between the declarative
265 judgment and the projection-based adapter; and a concrete-session realization theorem for
266 abstract witnesses. The 21-pack evaluation remains a proof of concept.

267 **Broader impact.** Pre-execution refutation reduces wasted computation (Appendix E) and prevents
268 structurally impossible agent plans from reaching deployment. The same formal verdict can, however,
269 create false confidence when an inaccurate pack omits a harmful tool effect or weakens the user’s
270 actual goal. We therefore expose the declared pack, witness or blocking frontier, and deferred
271 obligations rather than presenting a single trust score. Intent review and runtime payload monitoring
272 remain necessary before the method is used in consequential settings; this work does not evaluate
273 fairness, privacy, or deployment safety.

274 8 Conclusion

275 An LLM can draft an agent’s plan from intent; the open question has been whether anything can tell,
276 cheaply and before execution, that the plan is doomed. For the *achievability* question the answer is
277 yes, relative to a declared pack: combine planning’s capability-guarded reachability with a synthetic,
278 directly-typed global protocol; treat LLM compaction as untrusted; and use a refutation-sound
279 abstraction so an *impossible* verdict is never wrong about that declared model. *Achievable* remains
280 honest about payload and intent fidelity, while *Unknown* abstains outside static topology. This yields
281 a structured, inspectable verification signal for agent skills and a path toward, rather than a completed
282 proof of, stronger per-step safety.

283 References

- 284 [1] S.-S. Jongmans, F. Ferreira. Synthetic Behavioural Typing: Sound, Regular Multiparty Sessions via Implicit
285 Local Types. *ECOOP*, LIPIcs 263, 2023.
- 286 [2] D. Castro-Perez, F. Ferreira, S.-S. Jongmans. A Synthetic Reconstruction of Multiparty Session Types.
287 *POPL*, 2026.
- 288 [3] J. Li, X. Xiao, Y. Zhang, C. Liu, L. Zhao, X. Liao, Y. Ji, J. Wang, J. Gu, Y. Ge, W. Xu, X. Fang, X. Xu,
289 T. Zhao, Y. Kim, T. Wang, J. Hamm, S. Krishnaswamy, J. Huan, C. Reddy. Agent Harness Engineering: A
290 Survey. 2026.

- [4] M. Chen, J. Wang, Z. Liu, Y. Wang, H. Zheng, Q. Wang. From Failed Trajectories to Reliable LLM Agents: Diagnosing and Repairing Harness Flaws. arXiv:2606.06324, 2026.
- [5] Anthropic. Equipping Agents for the Real World with Agent Skills. Anthropic Engineering Blog, 16 Oct 2025; released as the open Agent Skills standard, <https://agentskills.io>, 18 Dec 2025.
- [6] S. Hu, C. Lu, J. Clune. Automated Design of Agentic Systems. *ICLR*, 2025.
- [7] J. Zhang, J. Xiang, Z. Yu, F. Teng, X. Chen, J. Chen, M. Zhuge, X. Cheng, S. Hong, J. Wang, et al. AFlow: Automating Agentic Workflow Generation. *ICLR*, 2025.
- [8] M. Cemri, M. Z. Pan, S. Yang, L. A. Agrawal, B. Chopra, R. Tiwari, K. Keutzer, A. Parameswaran, D. Klein, K. Ramchandran, M. Zaharia, J. E. Gonzalez, I. Stoica. Why Do Multi-Agent LLM Systems Fail? arXiv:2503.13657, 2025.
- [9] N. Yoshida, L. Gheri. A Very Gentle Introduction to Multiparty Session Types. *ICDCIT*, LNCS 11969, 2020.
- [10] S. Ghilezan, S. Jakšić, J. Pantović, A. Scalas, N. Yoshida. Precise Subtyping for Synchronous Multiparty Sessions. *J. Log. Algebr. Meth. Program.* 104, 2019.
- [11] A. Scalas, N. Yoshida. Less is More: Multiparty Session Types Revisited. *Proc. ACM Program. Lang.* 3(POPL), 2019.
- [12] K. Honda, N. Yoshida, M. Carbone. Multiparty Asynchronous Session Types. *J. ACM* 63(1), 2016.
- [13] S. Gay, M. Hole. Subtyping for Session Types in the Pi Calculus. *Acta Informatica* 42(2–3), 2005.
- [14] D. Brand, P. Zafiropulo. On Communicating Finite-State Machines. *J. ACM* 30(2), 1983.
- [15] E. Li, F. Stutz, T. Wies, D. Zufferey. Complete Multiparty Session Type Projection with Automata. *CAV* 2023.
- [16] R. Fikes, N. Nilsson. STRIPS: A New Approach to the Application of Theorem Proving to Problem Solving. *Artif. Intell.* 2(3–4), 1971.
- [17] B. Bollig, M. Függer, T. Nowak. Provable Coordination for LLM Agents via Message Sequence Charts. *ISoLA*, 2026. arXiv:2604.17612.
- [18] L. Miculicich, M. Parmar, H. Palangi, K. D. Dvijotham, M. Montanari, T. Pfister, L. T. Le. VeriGuard: Enhancing LLM Agent Safety via Verified Code Generation. arXiv:2510.05156, 2025.
- [19] V. P. Bhardwaj. Agent Behavioral Contracts: Formal Specification and Runtime Enforcement for Reliable Autonomous AI Agents. arXiv:2602.22302, 2026.
- [20] N. Paz, K. Pradeep, N. Raghavan, A. Nikirk, M. Gupta. SkillSpector: A Pre-Publication Security Control for Agent Skills. *Agent Skills Workshop at CAIS*, 2026.
- [21] NVIDIA. NVIDIA-Verified Agent Skills Provide Capability Governance for AI Agents. NVIDIA Technical Blog, 19 May 2026.

A Syntax

We fix disjoint sets of predicates $p \in \text{Pred}$ (boolean), numeric variables $x \in \text{Var}$, roles $p, q, r \in \mathcal{R}$, capability names $a \in \text{Cap}$, and message labels $\ell \in \text{Lab}$. Following the modern presentation of multiparty session types [9–11], we proceed bottom-up: state and capabilities (Sections A.1–A.2), then *processes* — the programs skills execute as — (Section A.3), then the global *type* they are checked against directly (Section A.4). In each grammar, “ $::=$ ” lists the possible forms of an expression. In each inference rule below, the statement beneath the line follows when every premise above the line holds. Rule names identify the corresponding checker obligation; they are not additional assumptions.

A.1 State logic

$$\begin{aligned}
 e & ::= x \mid n \mid e + e \mid e - e \mid c \cdot e & (c, n \in \mathbb{Z}) \\
 \varphi & ::= p \mid \top \mid \perp \mid \neg \varphi \mid \varphi \wedge \varphi \mid \varphi \vee \varphi \mid e \bowtie e & \bowtie \in \{<, \leq, =, \geq, >, \neq\}
 \end{aligned}$$

The state logic $\mathcal{L}$ is quantifier-free linear integer arithmetic with uninterpreted boolean predicates — a decidable fragment for which entailment $\models$ and satisfiability are discharged by a satisfiability-modulo-theories (SMT) solver.

336 A.2 Worlds and capabilities

337 A world $W = \langle B, N \rangle$ assigns truth values to predicates $B : \text{Pred} \rightarrow \mathbb{B}$ and numeric variables
338 $N : \text{Var} \rightarrow \mathbb{Z}$. A *capability context* Γ maps each available tool a to a guarded effect:

$$\begin{aligned} \Gamma(a) &= \langle pre_a, eff_a \rangle, \quad pre_a \in \mathcal{L}, \\ eff_a &= \langle Add_a \subseteq \text{Pred}, Del_a \subseteq \text{Pred}, Asg_a : \text{Var} \rightarrow e, Nd_a : \text{Var} \rightarrow \mathcal{L} \rangle \end{aligned}$$

339 $a \notin \text{dom}(\Gamma) \iff$ the agent does *not* possess tool a (no hallucinated tools).

340 Add/Del are the STRIPS add- and delete-lists; Asg gives deterministic numeric updates $x := e$; Nd
341 gives *nondeterministic* updates $x := *$ constrained by a formula over the new value (used to model
342 post-conditions such as “books a fare under 500”).

343 A.3 Processes and multiparty sessions

344 Skills *execute* as processes; a multi-agent run is a session of named processes. We use the synchronous
345 multiparty session calculus in its modern coinductive presentation [9, 10]: terms are *regular* trees
346 (finitely many distinct subtrees; $\mu X.P$ is notation for a cycle, identified with its unfolding), and labels
347 are already goal-relevant tokens, $\ell = \alpha(m)$ for concrete messages m — the tolerance dial α is a
348 parameter of the label alphabet.

$$P ::=^{coind} \mathbf{0} \mid p!\{\ell_i.P_i\}_{i \in I} \mid p?\{\ell_i.P_i\}_{i \in I} \mid a.P \quad \mathbb{M} = p_1[P_1] \parallel \cdots \parallel p_n[P_n]$$

349 with I finite and non-empty, labels ℓ_i pairwise distinct, and session names pairwise distinct. The
350 output $p!\{\ell_i.P_i\}$ *internally* chooses a label and sends it to p ; the input $p?\{\ell_i.P_i\}$ offers an *external*
351 choice; $a.P$ invokes capability a — the only construct that changes the world. A skill S_p is a process:
352 the behaviour declared for role p .

353 A.4 Global types

354 The single communication construct of a global type is a *directed* labelled choice — selection by p
355 and branching at q in one construct. (A partnerless “ p selects” admits no coherent view for the roles
356 that did not choose; directed choice is the standard MPST construct [12, 9].)

$$G ::=^{coind} p \rightarrow q : \{\ell_i.G_i\}_{i \in I} \mid a@p.G \mid \checkmark \varphi.G \mid \text{End}$$

357 subject to $p \neq q$ and the same regularity and label conditions as processes. This is the *only* type in the
358 discipline: Section C.2 types a whole session directly against G — there is no local-type grammar,
359 no projection function, and no separate subtyping relation. (A local-type projection of G remains a
360 legitimate, more efficient *algorithmic* decision procedure for the same conformance judgment, and is
361 what our reference implementation runs, Section 4; Section C.2 gives the specification it implements.)
362 Goal markers $\checkmark \varphi$ live *only* in G : achievability is decided on the global type (rule ACH), and markers
363 are consumed at typing time by T-GOAL without ever burdening a process. Only $a@p$ changes the
364 world; messages carry coordination information only.

365 B Operational Semantics

366 B.1 World transitions

A capability fires when its guard holds, producing a successor world. Because Nd is nondeterministic,
 $\langle W \rangle a \langle W' \rangle$ may relate one world to many. The relation is implicitly indexed by Γ : the notation
 pre_a, eff_a is defined only when $\Gamma(a) = \langle pre_a, eff_a \rangle$, so every firing presupposes $a \in \text{dom}(\Gamma)$.

$$\begin{array}{c} \text{WORLD-ACT} \\ W \models pre_a \quad W' \in \text{apply}(eff_a, W) \\ \hline \langle W \rangle a \langle W' \rangle \end{array}$$

367 $\text{apply}(eff_a, \langle B, N \rangle) = \langle B', N' \rangle, \quad B' = (B \cup Add_a) \setminus Del_a$

$$N'(x) = \begin{cases} \llbracket Asg_a(x) \rrbracket_N & \text{if } x \in \text{dom}(Asg_a) \\ v \text{ with } \langle B, N[x \mapsto v] \rangle \models Nd_a(x) & \text{if } x \in \text{dom}(Nd_a) \\ N(x) & \text{otherwise (frame)} \end{cases}$$

369 B.2 A labelled semantics for sessions and global types

370 Only capabilities touch the world, so sessions reduce as configurations $\langle \mathbb{M}; W \rangle$; communication is
 371 synchronous [10]. Each transition carries a label Λ recording its *participants*, so that independent
 372 moves by disjoint roles can be identified. Two auxiliary maps drive the labelled system: the
 373 *participants* $\text{prt}(\cdot)$ of a global type, a label, or a session, and the *capabilities* $\text{cap}(\cdot)$ (the set of moves
 374 a global type can perform). On global types,

$$\begin{aligned} \text{prt}(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}) &= \{p, q\} \cup \bigcup_{i \in I} \text{prt}(G_i) & \text{prt}(\check{\varphi}. G) &= \text{prt}(G) \\ \text{prt}(a @ p. G) &= \{p\} \cup \text{prt}(G) & \text{prt}(\text{End}) &= \emptyset, \end{aligned}$$

375 on labels $\text{prt}(p \ell q) = \{p, q\}$, $\text{prt}(a @ p) = \{p\}$, $\text{prt}(\check{\varphi}) = \emptyset$, and on sessions $\text{prt}(\mathbb{M}) = \{p \mid \mathbb{M} \equiv$
 376 $p[P] \ \& \ P \neq \mathbf{0}\}$ (the roles with a residual behaviour). The capabilities of a global type are

$$\begin{aligned} \text{cap}(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}) &= \{p \ell_i q \mid i \in I\} \cup \bigcup_{i \in I} \text{cap}(G_i) \\ \text{cap}(a @ p. G) &= \{a @ p\} \cup \text{cap}(G) & \text{cap}(\check{\varphi}. G) &= \{\check{\varphi}\} \cup \text{cap}(G) & \text{cap}(\text{End}) &= \emptyset. \end{aligned}$$

377 The capability names used by a protocol are

$$\text{caps}(G) = \{a \mid \exists p. a @ p \in \text{cap}(G)\}.$$

S-COMM

$$\frac{k \in I \subseteq J}{(p[q!\{\ell_i. P_i\}_{i \in I}] \parallel q[p?\{\ell_j. Q_j\}_{j \in J}] \parallel \mathbb{M}; W) \xrightarrow{p \ell_k q} (p[P_k] \parallel q[Q_k] \parallel \mathbb{M}; W)}$$

S-ACT

$$\frac{\langle W \rangle a \langle W' \rangle}{(p[a.P] \parallel \mathbb{M}; W) \xrightarrow{a @ p} (p[P] \parallel \mathbb{M}; W')}$$

S-GOAL

$$\frac{W \models \varphi}{(\mathbb{M}; W) \xrightarrow{\check{\varphi}} (\mathbb{M}; W)}$$

378 closed under permutation of $\parallel$ and $p[\mathbf{0}] \parallel \mathbb{M} \equiv \mathbb{M}$. A non-terminated configuration with no successor
 379 is *stuck*; the deadlocking planner/worker handoff of Section 1 is the stuck two-role configuration
 380 in which both processes begin with an input. A goal marker is *not* carried by any process; instead
 381 S-GOAL lets a configuration *observe* that its world already entails one of its declared goals φ , emitting
 382 the label $\check{\varphi}$ without altering $\mathbb{M}$ or W . (Each session carries a set of goal formulas, and φ ranges
 383 over that set — not over every formula the world happens to satisfy — so that each observation
 384 has a matching marker in the protocol.) This observable goal step is matched at the type level by
 385 G-GOAL-E below, so that $\check{\varphi}$ is a genuine label Λ of both transition systems and the operational
 386 correspondence of Section D.3 ranges over it uniformly.

The checker relates a session to its protocol through a *labelled* transition system on global types,
 $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, built from *head* rules that consume the leading construct and *interleaving*
 rules that let a move by participants *disjoint* from the head commute inward:

G-COMM-E

$$\frac{k \in I}{(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}, W) \rightsquigarrow_{p \ell_k q} (G_k, W)}$$

G-ACT-E

$$\frac{\langle W \rangle a \langle W' \rangle}{(a @ p. G, W) \rightsquigarrow_{a @ p} (G, W')}$$

G-GOAL-E

$$\frac{W \models \varphi}{(\check{\varphi}. G, W) \rightsquigarrow_{\check{\varphi}} (G, W)}$$

G-COMM-I

$$\frac{(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W') \ (\forall i \in I) \quad \Lambda \in \text{cap}(G_i) \quad \text{prt}(\Lambda) \cap \{p, q\} = \emptyset}{(p \rightarrow q : \{\ell_i. G_i\}_{i \in I}, W) \rightsquigarrow_{\Lambda} (p \rightarrow q : \{\ell_i. G'_i\}_{i \in I}, W')}$$

G-ACT-I

$$\frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G) \quad \text{prt}(\Lambda) \cap \{p\} = \emptyset}{(a @ p. G, W) \rightsquigarrow_{\Lambda} (a @ p. G', W')}$$

G-GOAL-I

$$\frac{(G, W) \rightsquigarrow_{\Lambda} (G', W') \quad \Lambda \in \text{cap}(G)}{(\check{\varphi}. G, W) \rightsquigarrow_{\Lambda} (\check{\varphi}. G', W')}$$

387 G-GOAL-E discharges a goal marker when $W \models \varphi$, emitting the observable label $\checkmark \varphi$ that matches
 388 S-GOAL. In the head-only reduction used for achievability, an unsatisfied marker is a checkpoint and
 389 blocks the path. In the full labelled correspondence, G-GOAL-I lets a continuation step commute
 390 inward under a still-pending marker. When that step changes the world, the correspondence theorem
 391 uses the goal-stability hypothesis stated in Section D.3; removing that hypothesis is future work. The
 392 extraction (head) rules G-COMM-E/G-ACT-E/G-GOAL-E give the Λ -erased reduction $(G, W) \rightsquigarrow$
 393 (G', W') that the achievability judgment below uses; the interleaving rules matter only for the
 394 operational correspondence of Section D.3, and their $\Lambda \in \text{cap}(G)$ premise confines a commuting
 395 move to a capability the residual protocol actually offers. G-COMM-E is legitimate because choice
 396 is directed — the branch is a communication both endpoints observe, and Section C.2 checks that
 397 every third party can follow. A well-formed pack uses its declared goal φ_{goal} at each goal marker. A
 398 configuration is *goal-satisfying* when control is at such a marker or at the terminus and the world
 399 entails that declared goal:

$$400 \quad \begin{aligned} & \text{goal}_{\varphi_{\text{goal}}}(\checkmark \varphi_{\text{goal}}.G, W) \iff W \models \varphi_{\text{goal}}, \quad \text{goal}_{\varphi_{\text{goal}}}(\text{End}, W) \iff W \models \varphi_{\text{goal}}, \\ & \Gamma; G \models \Diamond_{\text{init}} \varphi_{\text{goal}} \iff \exists W_0 \models \text{init}. \exists (G', W'). (G, W_0) \rightsquigarrow^* (G', W') \wedge \text{goal}_{\varphi_{\text{goal}}}(G', W'). \end{aligned}$$

401 The diamond is *existential*: a single witnessing run suffices — the formal source of detour-tolerance.

402 B.3 Realizability and subtyping, without projection

403 The classical route to ruling out a deadlocking handoff is *projection*: compute each role’s local
 404 type $G|r$ by structural recursion on G , taking the *merge* of a choice’s branches at every role that
 405 neither sends nor receives, and declaring G *realizable* exactly when this partial function is defined
 406 everywhere — undefinedness of the merge is the static signature of a role forced to behave differently
 407 in branches it cannot distinguish. Session subtyping $\leq$ is then a second, separate coinductive relation
 408 on local types, letting a role’s actual behaviour offer more external choices and fewer internal ones
 409 than its projected contract demands [13, 10, 11].

410 Both devices exist to answer questions about the *network as a whole* — “can every role tell the
 411 branches apart it needs to?”, “may this implementation stand in for that contract?” — while type-
 412 checking only one role’s syntax at a time. Section C.2 answers the same two questions with a single
 413 judgment, *typing the whole configuration directly against G* : a choice node’s rule quantifies over
 414 every branch the protocol declares and requires the *same* residual configuration to check against each
 415 one. This is the conservative plain-merge condition, obtained without ever computing $\sqcap$ or a local
 416 type. The receiver-side subtyping direction (a receiver may accept a superset of the protocol’s labels)
 417 is folded into the same rule as a label-set inclusion $I \subseteq J$. Tolerance therefore still has the same
 418 three independent knobs from Section 2 — the existential $\Diamond$ (Section B.2), subtyping (now inside
 419 T-COMM, Section C.2), and the granularity of α — “flexible, tolerant of small details” is still the
 420 profile $\langle \text{coarse } \alpha, \Diamond, \text{linear } \mathcal{L} \rangle$; a later safety layer flips the same machinery to the universal $\Box$ over
 421 permission predicates.

422 C The Achievability Type System

423 Achievability now factors into three premises, each separately decidable, combined by the top rule
 424 ACH: no hallucinated capability, direct conformance of the declared skills to G , and reachability of
 425 the goal.

426 C.1 Capability and goal typing

$$427 \quad \begin{array}{c} \text{T-EFF} \\ \Gamma(a) = \langle \text{pre}, \text{eff} \rangle \quad \varphi \models \text{pre} \\ \hline \Gamma \vdash \{\varphi\} a \{\text{sp}(\varphi, \text{eff})\} \end{array} \quad \begin{array}{c} \text{T-MISS} \\ a \notin \text{dom}(\Gamma) \\ \hline \Gamma \vdash a \triangleright \mathbf{X} \text{ (MISSING_CAPABILITY)} \end{array}$$

427 $\text{sp}(\varphi, \text{eff})$ is the strongest postcondition of the STRIPS effect. T-MISS fires on hallucinated planning:
 428 the plan names a tool the context does not grant, and achievability is refuted without any search.

429 C.2 Direct conformance typing

Skills are typed *directly* against the global protocol and the world, by the coinductive judgment $G \vdash$
 $(\mathbb{M}; W)$ — read “ G types session $\mathbb{M}$ starting from world W .” There is no local type, no projection,

and no separate subtyping relation: the receiver-side subtyping direction and the unobserved-choice check are structural side conditions of one rule.

$$\begin{array}{c}
\text{T-END} \\
\hline
\text{End} \vdash (p[0]; W)
\end{array}
\quad
\begin{array}{c}
\text{T-ACT} \\
\hline
G \vdash (p[P] \parallel \mathbb{M}; W') \quad \langle W \rangle a \langle W' \rangle \quad \text{prt}(G) \cup \{p\} = \text{prt}(\mathbb{M}) \cup \{p\} \\
\hline
a @ p. G \vdash (p[a.P] \parallel \mathbb{M}; W)
\end{array}$$

$$\begin{array}{c}
\text{T-GOAL} \\
\hline
G \vdash (\mathbb{M}; W) \quad W \models \varphi \quad \text{prt}(G) = \text{prt}(\mathbb{M}) \\
\hline
\checkmark \varphi. G \vdash (\mathbb{M}; W)
\end{array}$$

$$\begin{array}{c}
\text{T-COMM} \\
\hline
G_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}; W) \quad \forall i \in I \subseteq J \quad \text{prt}(p \rightarrow q : \{\ell_i.G_i\}_{i \in I}) = \text{prt}(\mathbb{M}) \cup \{p, q\} \\
\hline
p \rightarrow q : \{\ell_i.G_i\}_{i \in I} \vdash (p[q!\{\ell_i.P_i\}_{i \in I}] \parallel q[p?\{\ell_j.Q_j\}_{j \in J}] \parallel \mathbb{M}; W)
\end{array}$$

430 T-ACT pairs type and world in lockstep with WORLD-ACT: the unconsumed type $a @ p.G$ sits with
 431 the *pre*-effect world W and the residual G with the *post*-effect world W' — the same convention as
 432 G-ACT-E (Section B.2). In T-COMM the sender p offers exactly the protocol's branch set I , while
 433 the receiver q may accept *more* ($I \subseteq J$) — the one subtyping direction the rule keeps (SUB-EXT,
 434 safe interface slack at the receiver). Requiring *every* protocol branch ℓ_i to be checked against the
 435 *same* bystanders $\mathbb{M}$ is the plain-merge form of the unobserved-choice condition, so a role forced to
 436 behave differently across branches it cannot distinguish makes no derivation possible — the rule fails
 437 closed, and deadlock-freedom falls out of T-COMM itself with no merge to compute (Theorem 4).
 438 Each rule additionally carries a *participant-matching* side condition tying the protocol's participants
 439 to the session's: $\text{prt}(p \rightarrow q : \{\ell_i.G_i\}) = \text{prt}(\mathbb{M}) \cup \{p, q\}$ in T-COMM, $\text{prt}(G) \cup \{p\} = \text{prt}(\mathbb{M}) \cup \{p\}$
 440 in T-ACT, and $\text{prt}(G) = \text{prt}(\mathbb{M})$ in T-GOAL. These make the invariant “the roles the protocol still
 441 mentions are exactly the roles still running” hold at every node, ruling out a session that carries a stray
 442 participant the protocol has forgotten (or vice versa) — the well-formedness that lets the interleaving
 443 cases of Section D.3 match a bystander move on the session to a $\text{cap}(\cdot)$ -labelled move on the type.
 444 T-END closes the discipline at the terminus: the terminated protocol End types a single idle role $p[0]$
 445 — and, through the congruence $p[0] \parallel \mathbb{M} \equiv \mathbb{M}$, any session all of whose roles have terminated —
 446 the base case of the participant invariant, since $\text{prt}(\text{End}) = \emptyset$ and an idle role contributes nothing to
 447 $\text{prt}(\mathbb{M})$. T-ACT is derivable only when $\langle W \rangle a \langle W' \rangle$ holds, which itself presupposes $a \in \text{dom}(\Gamma)$:
 448 this is where hallucinated capabilities die, with T-MISS as the diagnostic.

449 C.3 The achievability judgment

$$\begin{array}{c}
\text{ACH} \\
\hline
\text{caps}(G) \subseteq \text{dom}(\Gamma) \\
\exists W_0 \models \text{init}. G \vdash (p_1[S_{p_1}] \parallel \dots \parallel p_n[S_{p_n}]; W_0) \wedge \Gamma; (G, W_0) \models \Diamond \varphi_{\text{goal}} \\
\hline
\Gamma \vdash \{S_p\}_p : G \triangleright \Diamond \varphi_{\text{goal}}
\end{array}$$

450 Read top-to-bottom: no hallucinated tools; some plausible initial world exists from which the declared
 451 skills directly conform to the protocol (deadlock-free, every capability call guarded) *and* that same
 452 world reaches the goal. The reachability half is unchanged from Section B.2 and decided on Γ, G
 453 alone, before any S_p is known — exactly what lets the checker run on the LLM-compacted pack
 454 the moment it exists; conformance is the additional, separately decidable check once the skills are
 455 declared, and needs no transport lemma: the receiver-side subtyping slack is already inside T-COMM.

456 C.4 Reachability rules

The diamond is derived by the following inductive system, realized by the checker as a finite forward search over $\rightsquigarrow$; unlike T-COMM/T-ACT/T-GOAL, it never mentions a session $\mathbb{M}$, which is what makes it decidable on the pack alone.

$$\begin{array}{c}
\text{REACH-GOAL} \\
\hline
W \models \varphi_{\text{goal}} \quad G = \text{End} \vee \exists G'. G = \checkmark \varphi_{\text{goal}}. G' \\
\hline
\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}
\end{array}$$

$$\begin{array}{c}
\text{REACH-STEP} \\
\hline
(G, W) \rightsquigarrow (G', W') \quad \Gamma; (G', W') \models \Diamond \varphi_{\text{goal}} \\
\hline
\Gamma; (G, W) \models \Diamond \varphi_{\text{goal}}
\end{array}
\quad
\begin{array}{c}
\text{REACH-OR} \\
\hline
\exists k \in I. \Gamma; (G_k, W) \models \Diamond \varphi_{\text{goal}} \\
\hline
\Gamma; (p \rightarrow q : \{\ell_i.G_i\}_{i \in I}, W) \models \Diamond \varphi_{\text{goal}}
\end{array}$$

457 D Metatheory

458 We separate what is *mechanized in Coq* (Theorems 1–3; Theorems 4–5 for the single-label com-
 459 munication fragment with full interleaving; the head-move action case; and the concrete instances,
 460 all axiom-free) from what is *proved on paper* (the action-interleaving cases, decidability, and the
 461 undecidability boundary).

462 D.1 Soundness of the abstraction

463 Let $\mathcal{C}$ be the concrete configuration space of protocol/world pairs, $\mathcal{A}$ the abstract configuration space
 464 the checker explores, and $abs : \mathcal{C} \rightarrow \mathcal{A}$ the abstraction. We require:

$$(step\text{-}sim) \ C \rightsquigarrow C' \Rightarrow abs(C) \hat{\rightsquigarrow} abs(C'), \quad (goal\text{-}sim) \ goal_{\varphi_{goal}}(C) \Rightarrow \hat{goal}(abs(C)).$$

465 **Lemma 1** (Transport). *If $C_0 \rightsquigarrow^* C$ then $abs(C_0) \hat{\rightsquigarrow}^* abs(C)$.*

466 *Proof.* By induction on the reflexive-transitive closure. The empty run is reflexivity. For $C_0 \rightsquigarrow^*$
 467 $C' \rightsquigarrow C$, the induction hypothesis gives $abs(C_0) \hat{\rightsquigarrow}^* abs(C')$ and STEP-SIM gives $abs(C') \hat{\rightsquigarrow} abs(C)$;
 468 compose them. Mechanized as `reach_abs`, parameterized by STEP-SIM. $\square$

469 **Lemma 2** (Reachability monotonicity). *If every step of $\rightsquigarrow_1$ is a step of $\rightsquigarrow_2$ (that is, $x \rightsquigarrow_1 y$ implies
 470 $x \rightsquigarrow_2 y$), then $s \rightsquigarrow_1^* w$ implies $s \rightsquigarrow_2^* w$.*

471 *Proof.* By induction on the closure $s \rightsquigarrow_1^* w$. The empty run transports by reflexivity. For $s \rightsquigarrow_1^* u \rightsquigarrow_1$
 472 w , the induction hypothesis gives $s \rightsquigarrow_2^* u$; the hypothesis turns the last step $u \rightsquigarrow_1 w$ into $u \rightsquigarrow_2 w$;
 473 appending it gives $s \rightsquigarrow_2^* w$. Mechanized as `reach_mono`. $\square$

474 D.2 Refutation soundness, tolerance, monotonicity

475 **Theorem 1** (Refutation soundness). *If the abstract system has no reachable goal state from
 476 $abs(G, W_0)$, then no declared run of that pack configuration reaches φ_{goal} . Hence an impossi-
 477 ble verdict is never wrong relative to the pack, provided STEP-SIM and GOAL-SIM hold.*

478 *Proof.* We prove the contrapositive: if *some* declared run reaches the goal, then the abstract
 479 system reaches an abstract goal state. Suppose $C_0 \rightsquigarrow^* C$ with $goal_{\varphi_{goal}}(C)$. By Lemma 1,
 480 $abs(C_0) \hat{\rightsquigarrow}^* abs(C)$, and by GOAL-SIM, $\hat{goal}(abs(C))$. Thus $abs(C)$ is an abstract goal state reach-
 481 able from $abs(C_0)$, contradicting the hypothesis. Hence no declared run reaches φ_{goal} : a verdict
 482 of *impossible* — read off exactly the premise, that the abstract search exhausts without hitting an
 483 abstract goal — is never wrong. Mechanized, axiom-free as a theorem schema parameterized by the
 484 two simulation hypotheses (the existential witness is $abs(C)$): $\square$

```
485 Theorem refutation_sound : forall s0,
486   (~ exists a, reach_astype (abs s0) a /\ agoal a) ->
487   (~ exists w, reach_cstep s0 w /\ cgoal w).
488   (* Print Assumptions refutation_sound. => Closed under the global
489   context *)
```

490 **Theorem 2** (Tolerance soundness). *If a coarser over-approximation (more edges) cannot reach the
 491 goal, neither can any finer system. Hence coarsening α to tolerate more detail never produces a false
 492 refutation.*

493 *Proof.* Let $\rightsquigarrow_{fine}$ and $\rightsquigarrow_{coarse}$ be the two abstract step relations, with the coarsening an *over-*
 494 approximation: every fine step is also a coarse step, $x \rightsquigarrow_{fine} y \Rightarrow x \rightsquigarrow_{coarse} y$. Suppose, for
 495 contradiction, that the finer system reaches the goal: $s_0 \rightsquigarrow_{fine}^* a$ with $\hat{goal}(a)$. By Lemma 2 (with
 496 $\rightsquigarrow_1 = \rightsquigarrow_{fine}$, $\rightsquigarrow_2 = \rightsquigarrow_{coarse}$), $s_0 \rightsquigarrow_{coarse}^* a$; and $\hat{goal}(a)$ still holds. So the coarser system reaches
 497 a goal state, contradicting the hypothesis. Hence no coarser-refuted goal is reachable in any finer
 498 system: coarsening α only *adds* abstract edges, so it can never turn an unreachable goal into a
 499 reachable one — refutation is preserved. Mechanized as `tolerance_sound`, axiom-free. $\square$

Theorem 3 (Capability monotonicity). *If $\Gamma \subseteq \Gamma'$ and Γ reaches φ_{goal} , then Γ' reaches φ_{goal} . Granting tools never makes an achievable goal impossible.*

Proof. Enlarging the capability context only *adds* guarded effects: since $\Gamma \subseteq \Gamma'$, every capability firing available under Γ is available under Γ' with the same guard and effect (WORLD-ACT quantifies over $a \in \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$), so every step of the Γ -transition relation is a step of the Γ' -relation. Suppose Γ reaches the goal configuration C from C_0 . By Lemma 2, the same configuration is reachable under Γ' , and $\text{goal}_{\varphi_{goal}}(C)$ is unchanged because the goal predicate does not depend on Γ . Hence Γ' reaches φ_{goal} via the same witnessing run. The other premises of ACH are monotone in the same direction ($\text{caps}(G) \subseteq \text{dom}(\Gamma) \subseteq \text{dom}(\Gamma')$ still holds, and realizability and conformance do not mention Γ beyond $a \in \text{dom}(\Gamma)$), so the whole judgment is preserved. The transport half is mechanized as `cap_monotone`, axiom-free, stated over an inclusion of step relations; the reduction of $\Gamma \subseteq \Gamma'$ to that inclusion is the WORLD-ACT argument above, on paper. $\square$

Concrete instance. The flight skill of Section 2, variant A, is mechanized as `FlightInstance`. We prove `flight_refuted` (no run reaches `booked` $\wedge$ `confirmation_sent`, since no step touches `conf`) and `booking_reachable` (a booking is still reachable). Both are axiom-free, witnessing non-vacuity of Theorem 1.

D.3 Operational correspondence: subject reduction and session fidelity

The direct judgment $G \vdash (\mathbb{M}; W)$ is not merely preserved by reduction: the session and its protocol step *in lockstep* over the labelled transition systems of Section B.2. This is the standard soundness backbone of a session calculus [12, 11], obtained here *directly*, with no projection and no merge. For world-changing interleavings, the statements below assume two explicit frame conditions: effects of participant-disjoint actions commute, and an action moved under a pending marker preserves that marker's formula. Communication-only reductions satisfy both conditions vacuously.

Lemma 3 (Residual capability). *If $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, then $\Lambda \in \text{cap}(G)$.*

Lemma 4 (Participant agreement). *If $G \vdash (\mathbb{M}; W)$, then $\text{prt}(G) = \text{prt}(\mathbb{M})$.*

Both follow by induction on the transition or typing derivation, respectively; they are the capability-membership and participant invariants used in the interleaving cases below.

Theorem 4 (Subject reduction). *If $G \vdash (\mathbb{M}; W)$ and $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$, then $(G, W) \rightsquigarrow_{\Lambda} (G', W')$ for some G' with $G' \vdash (\mathbb{M}'; W')$.*

Theorem 5 (Session fidelity). *If $G \vdash (\mathbb{M}; W)$ and $(G, W) \rightsquigarrow_{\Lambda} (G', W')$, then $(\mathbb{M}, W) \xrightarrow{\Lambda} (\mathbb{M}', W')$ for some $\mathbb{M}'$ with $G' \vdash (\mathbb{M}'; W')$.*

We prove both by induction on the typing derivation $G \vdash (\mathbb{M}; W)$, with a case analysis on the transition; the one auxiliary fact used throughout is that typing respects pointwise-equal sessions (`ctypes_ext`), which lets the interleaving cases reassociate independent updates without functional extensionality.

Proof of Theorem 4 (subject reduction). By induction on $G \vdash (\mathbb{M}; W)$.

Case T-END ($G = \text{End}$, every role idle). Then $\mathbb{M}$ has no output or pending action, so the only available transition is the goal observation S-GOAL at $\Lambda = \checkmark \varphi_{goal}$ when $W \models \varphi_{goal}$, which leaves $(\mathbb{M}, W)$ fixed; there is no residual type to preserve and the claim holds (otherwise no Λ -transition exists and it holds vacuously).

Case T-GOAL ($G = \checkmark \varphi.G_0$, with $W \models \varphi$ and $G_0 \vdash (\mathbb{M}; W)$, $\text{prt}(G_0) = \text{prt}(\mathbb{M})$). The transition $(\mathbb{M}, W) \rightarrow_{\Lambda} (\mathbb{M}', W')$ splits on its label.

- **Goal observation** ($\Lambda = \checkmark \varphi$, by S-GOAL, so $\mathbb{M}' = \mathbb{M}$ and $W' = W$). Since $W \models \varphi$, rule G-GOAL-E discharges the marker, $(\checkmark \varphi.G_0, W) \rightsquigarrow_{\checkmark \varphi} (G_0, W)$, and $G_0 \vdash (\mathbb{M}; W)$ is the premise: take $G' = G_0$.
- **Interleaving** ($\Lambda \neq \checkmark \varphi$). A goal marker is not a session construct, so this is already a transition of G_0 's session; the induction hypothesis gives $(G_0, W) \rightsquigarrow_{\Lambda} (G', W')$ with

547 $\Lambda \in \text{cap}(G_0)$ and $G' \vdash (\mathbb{M}'; W')$. When the move is world-preserving (a communica-
 548 tion, $W' = W$), G-GOAL-I carries the still-pending marker along, $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda}$
 549 $(\check{\varphi}.G', W)$, and $\check{\varphi}.G' \vdash (\mathbb{M}'; W)$ by T-GOAL (the side condition $\text{prt}(G') = \text{prt}(\mathbb{M}')$ is
 550 preserved because Λ acts on the same participants on both sides). When the move is *world-*
 551 *changing* (a capability firing, $W' \neq W$), G-GOAL-I still commutes it under the marker,
 552 $(\check{\varphi}.G_0, W) \rightsquigarrow_{\Lambda} (\check{\varphi}.G', W')$; re-typing the residual by T-GOAL then additionally asks
 553 $W' \models \varphi$. This follows from the goal-stability hypothesis on actions commuting under pend-
 554 ing markers. The global type sequences the marker before G_0 's actions whereas the session
 555 may fire the action first; the interleaving rule lets the type catch up under that hypothesis.
 556 Removing the hypothesis, rather than this conditional case, is the open correspondence
 557 problem recorded in Section 7.

558 *Case T-COMM*, with head $p \rightarrow q$ and branches G_i . Here $\mathbb{M}$ is $p[q! \dots] \parallel q[p? \dots] \parallel \mathbb{M}_0$, and each
 559 branch satisfies $G_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0; W)$. If the transition is a goal observation $\Lambda = \check{\varphi}$
 560 (S-GOAL, leaving $(\mathbb{M}, W)$ fixed), it is matched by G-COMM-I: $\text{prt}(\check{\varphi}) = \emptyset$ makes the disjointness
 561 premise trivial, and each branch realises the same observation by the induction hypothesis, using
 562 $\check{\varphi} \in \text{cap}(G_i)$ — the observation concerns a marker the protocol actually pends (see the note below).
 563 Otherwise the transition is a communication $\Lambda = r \ell t$. Because p offers only outputs to q and q only
 564 inputs from p , the sender/receiver shapes force a dichotomy.

- 565 • *Head move* ($r = p$). Then $t = q$ and $\ell = \ell_k \in I$, and $\mathbb{M}' = p[P_k] \parallel q[Q_k] \parallel \mathbb{M}_0$. Rule
 566 G-COMM-E gives $(G, W) \rightsquigarrow_{p\ell_k q} (G_k, W)$, and $G_k \vdash (\mathbb{M}'; W)$ is exactly the k -th premise.
- 567 • *Interleaving move* ($r \neq p$). The shapes then force $r, t \notin \{p, q\}$: r is an output so $r \neq q$,
 568 and t is an input so $t \neq p$, while $t = q$ would force $r = p$. Since r, t lie in the bystanders
 569 $\mathbb{M}_0$, the *same* communication is enabled in each branch's configuration $p[P_i] \parallel q[Q_i] \parallel \mathbb{M}_0$
 570 (the head updates do not touch r, t). The induction hypothesis at each i yields G'_i with
 571 $(G_i, W) \rightsquigarrow_{\Lambda} (G'_i, W)$ and $G'_i \vdash (p[P_i] \parallel q[Q_i] \parallel \mathbb{M}'_0; W)$, where $\mathbb{M}'_0$ is $\mathbb{M}_0$ after the
 572 bystander step. As $\text{prt}(\Lambda) = \{r, t\}$ is disjoint from $\{p, q\}$, rule G-COMM-I assembles
 573 $(G, W) \rightsquigarrow_{\Lambda} (p \rightarrow q: \{\ell_i. G'_i\}, W)$, and re-applying T-COMM (the sender/receiver at p, q
 574 are untouched, and each branch is retyped by the induction hypothesis, reassociating the
 575 independent updates via `ctypes_ext`) gives the residual typing.

576 Communications leave the world fixed, so no independence hypothesis is needed. For a world-
 577 changing action at the head (T-ACT), G-ACT-E matches and the residual G is typed at the *same*
 578 post-effect world the rule reaches — this is exactly where the lockstep world pairing of the corrected
 579 T-ACT is used; a goal observation at an action-typed configuration is matched by G-ACT-I exactly
 580 as in the T-COMM case, since $\text{prt}(\check{\varphi}) \cap \{p\} = \emptyset$ trivially. Interleaving *two* actions requires their
 581 effects to commute ($\text{eff}_a \circ \text{eff}_b = \text{eff}_b \circ \text{eff}_a$ for disjoint a, b), a frame-independence side condition
 582 that participant-disjointness alone does not supply; under it the T-ACT interleaving case goes through
 583 identically. $\square$

584 *Proof of Theorem 5 (session fidelity)*. By induction on $G \vdash (\mathbb{M}; W)$, inverting the global step
 585 $(G, W) \rightsquigarrow_{\Lambda} (G', W')$. T-END admits only the goal observation G-GOAL-E at $\check{\varphi}_{\text{goal}}$, matched
 586 on the session by S-GOAL. For T-GOAL, the step is either G-GOAL-E — matched by S-GOAL
 587 on the session, the marker discharged — or G-GOAL-I, whose premise $(G_0, W) \rightsquigarrow_{\Lambda} (G'_0, W')$
 588 the induction hypothesis realises as an underlying session step that carries the marker along. For
 589 T-COMM, the global step is either G-COMM-E — and the prescribed head communication $p \ell_k q$ is
 590 enabled by S-COMM, landing in the k -th premise — or G-COMM-I with $\text{prt}(\Lambda) \cap \{p, q\} = \emptyset$; then
 591 the induction hypothesis on any branch produces a bystander communication, whose endpoints lie
 592 outside $\{p, q\}$, so S-COMM fires it on $\mathbb{M}$ and T-COMM retypes the result. The action cases mirror
 593 those of Theorem 4. $\square$

594 **Mechanization.** Both theorems are mechanized axiom-free for the single-label communication
 595 fragment — the fragment in which bystander interleaving is unconditional — in `DirectTypingSR.v`,
 596 over labelled session and global transition systems, with no projection, merge, or local type:

```
597 Theorem subject_reduction : forall W G s s' L,
598   ctypes W G s -> lstep L s s' ->
```

```

599     exists G', gstep W L G G' /\ ctypes W G' s'.
600     Theorem session_fidelity : forall W G G' s L,
601       ctypes W G s -> gstep W L G G' ->
602       exists s', lstep L s s' /\ ctypes W G' s'.
603     (* Print Assumptions => Closed under the global context (both). *)

```

604 The head-move case for world-changing *actions* is additionally mechanized in `DirectTyping.v`
605 (type_directed_safety, also progress). Observable goal labels appear in neither Coq develop-
606 ment: the mechanized label alphabet is communication-only, and its goal rule fuses marker discharge
607 with a continuation step. Participant-matching side conditions, labelled action interleaving, and
608 multi-branch choice are likewise absent. Completing a faithful mechanization of the paper rules
609 remains work in Section 7.

610 **Concrete instance.** `HandoffInstance` mechanizes the planner/worker handoff of Section 1 on both
611 sides. The *good* pairing — planner sends, worker delivers, goal checked — is proved `ctypes`-typed
612 (`good_typed`) and its run is proved to reach a world satisfying the goal (`good_runs_to_goal`). The
613 *bad* pairing — both roles start with an input, exactly the deadlock of Section 1 — is proved `Stuck`
614 (`bad_stuck`), and the contrapositive of progress then gives, for free, that *no non-trivial global type*
615 *can ever type it* (`bad_untypable`): the rule rejects the classical deadlocking handoff without any
616 projection ever being computed.

617 D.4 Incompleteness, by design

618 **Proposition 1** (Incompleteness). $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ may hold while no concrete runtime execution
619 reaches φ_{goal} : the witnessing abstract path can be spurious when α collapses a distinction that
620 mattered. The system is sound for $\times$ and incomplete for $\checkmark$.

621 This is the price of decidable, tolerant over-approximation; tightening α trades tolerance for precision.
622 The residue is confined to two edges, motivating the layered architecture:

- 623 • **Payload faithfulness** (bottom edge): a tool’s declared post-condition (“filters to fares
624 < 500”) may be false — a runtime obligation for a deterministic monitor (Layer C), not a
625 static one.
- 626 • **Intent fidelity** (top edge): the formalized goal may not capture what the user meant —
627 irreducibly semantic, surfaced at the single compaction checkpoint for human review.

628 D.5 Decidability and the autonomy boundary

629 **Theorem 6** (Decidability). *In the fragment with finitely many roles (static topology), regular global*
630 *types represented by finite tail-recursive control graphs, decidable state logic $\mathcal{L}$, and $\mathcal{L}$ -definable*
631 *effects, $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ is decidable.*

632 *Proof.* The pack-level achievability judgment $\Gamma; G \models \Diamond_{init} \varphi_{goal}$ (Section B.2) is an existential
633 reachability query over configurations (G', W) ; we exhibit a terminating decision procedure and
634 argue its completeness.

635 *The search space is finite.* A configuration has two components. (i) *Control*, the residual global
636 type G' reachable from G by $\rightsquigarrow$. Since G is a regular tree (tail-recursion: finitely many distinct
637 subtrees, Section A.3) and each $\rightsquigarrow$ step either descends into a subtree (G-COMM-E, G-ACT-E) or
638 erases a marker (G-GOAL-E), the set $\{G' : (G, _) \rightsquigarrow^* (G', _)\}$ is a subset of the finitely many
639 subtrees of G , hence finite. (ii) *Symbolic world*. The boolean part is a valuation $B : \text{Pred} \rightarrow \mathbb{B}$
640 over the finitely many predicates named in the pack, so there are $2^{|\text{Pred}|}$ boolean states. The numeric
641 part is summarized by an accumulated constraint $\psi \in \mathcal{L}$ (a conjunction of the guards taken and
642 the Nd -postconditions assumed); on a control-graph back edge the reference procedure applies the
643 widening $\psi \mapsto \top$, forgetting accumulated numeric constraints. This one-element numeric summary
644 is an over-approximation and, together with the finite boolean valuations, leaves only finitely many
645 symbolic worlds. The product with finite control is therefore finite.

646 *Each edge is computable.* Firing a capability requires checking $W \models pre_a$ and forming sp ; testing
647 goal-satisfaction requires $W \models \varphi$. Both are entailment/satisfiability queries in quantifier-free linear
648 integer arithmetic with uninterpreted predicates — a decidable theory, discharged by an SMT solver.

Branch selection (REACH-OR) is a finite disjunction. Existence of an initial world is decided by the same SMT query over *init*; each satisfying symbolic initial valuation seeds the finite search.

Termination and correctness. Forward search (breadth-first over $\rightsquigarrow$) maintains a visited set of (control, symbolic-world) pairs; because that set is finite it adds each pair at most once, so the search halts. It answers ACHIEVABLE iff it enumerates a goal-satisfying configuration, and IMPOSSIBLE otherwise. Soundness of the IMPOSSIBLE verdict is Theorem 1; and widening only *enlarges* the reachable set (it weakens constraints, adding edges), so by Theorem 2 it never introduces a false refutation. Hence the procedure is a decision procedure for $\Gamma; G \models \Diamond_{init} \varphi_{goal}$. $\square$

Proposition 2 (Undecidability in a dynamic-topology extension). *Consider the extension G^+ of the grammar with spawn $r.G$, which creates fresh participants at run time. With an unbounded role population and dynamic channels, achievability in G^+ is undecidable.*

Proof sketch. By reduction from the control-state reachability problem for communicating finite-state machines (CFSMs), which is undecidable [14]. Fix a CFSM instance: finite-control machines M_1, M_2 exchanging messages over unbounded FIFO channels, and a target control state q_f of M_1 ; deciding whether a run reaches q_f is undecidable.

We compute, from this instance, a skill whose goal is achievable iff the CFSM reaches q_f . Each machine M_i becomes a role whose process mirrors its control graph: a transition that sends message m spawns, at run time, a fresh *courier* participant carrying m (this is exactly the dynamic-spawning capability the theorem hypothesizes), and a transition that receives m synchronizes with the oldest outstanding courier for that channel, so the unbounded family of couriers realizes the unbounded FIFO. The required order can be made explicit by assigning sequence numbers through the numeric world state and guarding receives on the next sequence number; the *Asg/Nd* machinery of Section A.2 supplies those updates. A boolean predicate at_q_f is established by the single capability guarding M_1 ’s entry into q_f , and we set $\varphi_{goal} = at_q_f$. By construction a run of the skill reaches a state satisfying φ_{goal} iff the CFSM has a run reaching q_f : the courier discipline then puts the skill’s reachable configurations in correspondence with the CFSM’s channel contents and control states. The construction is a computable map from CFSM instances to packs, so a decision procedure for achievability would decide CFSM control-state reachability — impossible. Hence achievability is undecidable once participants may be spawned dynamically. (The finiteness argument of Theorem 6 breaks at exactly the point this reduction exploits: an unbounded number of couriers makes the set of reachable controls infinite.) $\square$

The two results isolate one concrete boundary: static finite topology admits a total decision procedure, while unbounded dynamic spawning does not. This is a boundary on topology, not a claim that all forms of agent autonomy are undecidable. The implementation returns UNKNOWN when a pack violates the static-topology side condition of Theorem 6, unless an independent capability or establisher refutation has already proved it impossible.

D.6 The partition of obligations

Corollary 1. *The architecture separates “will the goal be achieved” into four obligations on disjoint substrates:*

Concern	Where	Mechanism
goal achievability	static, decidable	Coq specification; deterministic search; no LLM
per-step safety / least privilege	Layer B (future)	same engine, $\square$ + permission tiers
payload faithfulness	Layer C (future), run-time	deterministic monitors with blame
intent fidelity	top edge, synthesis	one human checkpoint

D.7 Verifiable feedback beyond a scalar reward

The verdict is a heterogeneous verification signal rather than a scalar score. An IMPOSSIBLE result identifies which premise failed and carries a blocking frontier; ACHIEVABLE carries a witness path and explicitly retains the two deferred obligations; UNKNOWN is an abstention outside the decidable fragment, not a refutation. These structured outcomes can serve as cheap negative labels

for automated skill or prompt search without an LLM judge, while the intent-fidelity checkpoint keeps the irreducible semantic decision with a human. We treat such search as an application of the verifier, not as a contribution evaluated here.

E Token economics of pre-execution refutation

The practical case for deciding achievability *before* execution is an economic one, so we state it in tokens rather than adjectives. Two asymmetries carry it. First, no model sits in the trusted path: the schema gate, projection, conformance and Z3 reachability spend **zero tokens**, and the deterministic front-end spends zero as well. Only the optional LLM compaction spends tokens, and it is paid *once per skill version*. Second, an agent turn re-sends its whole context: writing S for the harness prompt, K for the loaded skill, g for the mean tokens appended per turn and o for output per turn, a run of T turns reads $T(S+K) + gT(T-1)/2$ input and To output tokens. Compaction is linear in the skill and paid once; a doomed run is *quadratic* in its turns and recurs on every invocation.

Runtime waste is necessarily a model — it prices a run that, if the refutation is correct, never happens — so we publish the model rather than a single number. Each refutation reason carries a (low, typical, high) band for how long an agent flails before that failure stops it, and how many agents are billed. With conservative defaults ($S=2500$, $g=700$, $o=250$) and a median real consumer skill ($K \approx 1.7\text{K}$ tokens), compaction costs $\sim 2.8\text{K}$ tokens against the following avoided waste per prevented invocation:

Verdict reason	turns (lo–typ–hi)	waste/run	leverage
MISSING_CAPABILITY	3–8–14, 1 agent	50 240	18×
GOAL_UNSAT	8–18–30, 1 agent	176 040	63×
NON_CONFORMANT	6–15–30, 2 agents	261 900	94×
NON_PROJECTABLE	6–15–30, 2 agents	261 900	94×
BLOCKED_GUARD	12–25–50, 1 agent	305 750	110×

The ordering is the robust part, and it is the expected one. `MISSING_CAPABILITY` is cheapest at run time: the agent calls a tool that is not there and learns immediately, and most of the cost is the improvisation that follows. `BLOCKED_GUARD` is dearest, because a precondition no run can satisfy is precisely the case an agent cannot learn from — it retries to the harness turn cap. The two multi-party reasons bill two contexts. `GOAL_UNSAT` additionally carries a cost the token bill does not show: a run whose goal conjunct has no establisher can terminate *believing* it succeeded. `DYNAMIC_TOPOLOGY` claims no savings at all, because an abstention prevents nothing.

Over the 15-spec corpus the seven structural refutations cost 12.2K tokens of compaction and avoid $\sim 1.07\text{M}$ tokens per round of invocations (band 0.28M–3.15M): $\sim 87\times$ leverage, or nothing at all on the deterministic path. Break-even therefore falls *inside* the first prevented run in every mode — between 0.9% and 5.5% of it. The honest denominator for a healthy skill, where the check buys nothing and still costs something, is 2.9% of one successful run over the corpus and 4.0% for a median real skill; with the deterministic front-end, zero. Prompt caching changes the price of the wasted tokens, not their number. Halving every waste default leaves leverage at $9\times$ – $55\times$ and the ordering unchanged. The model, its defaults and their justification are in the artifact (`skillc.tokens`, `skillc.cost`); the Coq development costs no tokens and is amortized over every skill ever checked.

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744 a complete (and correct) proof?

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